

AI Support Agent for AppleSupport (Twitter)

Evaluation Report — Customer Support on Twitter Dataset

Name: Manda Sanjana

Email: sanjanakiran19@gmail.com

GitHub Repository: <https://github.com/mandasanjana19/support-agent>

1 Problem Framing

We built an AI support agent for AppleSupport, one brand handle within the Kaggle Customer Support on Twitter dataset. The agent takes an incoming customer tweet and (1) classifies it into one of eight hand-defined intents, (2) drafts a reply grounded in retrieved historical (customer message, brand reply) pairs from AppleSupport’s real conversation history, and (3) decides whether the message can be auto-handled or must be escalated to a human, with a stated reason.

1.1 Why AppleSupport

We compared AppleSupport against AmazonHelp during brand selection. AmazonHelp’s replies are dominated by generic “please DM us your order number” deflections, which gives a retrieval-grounded agent very little real resolution text to learn from. AppleSupport’s threads, by contrast, contain concrete troubleshooting language (iOS update failures, battery drain, Apple ID lock-outs) and specific resolution steps stated directly in-thread. This makes grounded reply drafting meaningfully more testable.

1.2 What “good” means for this brand

- **Correct intent classification:** the agent should route a message to the same category a careful human labeler would choose, often disambiguating between closely related categories (e.g. battery drain that started after an update).
- **Grounded, non-hallucinatory replies:** replies should reflect real resolution patterns seen in this brand’s history — not invent specific refund amounts, timelines, or policies never observed in the data.
- **Conservative, explainable escalation:** cases with security, legal, or financial risk signals should escalate even at the cost of some false positives, and every routing decision must carry a plain-language reason a human reviewer can audit.

1.3 What we chose not to build

- Multi-turn conversation context beyond a single (customer message, brand reply) pair.
- A fine-tuned classifier — intent classification uses a prompted LLM (Groq/Llama) plus a TF-IDF + Logistic Regression baseline for comparison.
- Embedding-based semantic retrieval — grounding uses TF-IDF cosine similarity instead of sentence embeddings, a deliberate dependency-weight tradeoff (see Decision Log).
- Full-dataset training — we work from a ~3,000-pair subsample of AppleSupport’s 106,000+ tweets.
- Production deployment, authentication, or a live Twitter API integration.

2 Methodology

2.1 Intent Taxonomy

We discovered candidate intents by TF-IDF + KMeans clustering ($k = 8$) over AppleSupport customer messages, inspected the top terms and sample messages per cluster, then hand-wrote a final taxonomy of eight brand-specific intents:

```
system_update_issue,  
battery_performance,  
software_bug_glitch,  
apple_id_account,  
app_store_media,  
device_hardware_connectivity,  
complaint_no_specific_ask, and  
other.
```

Note: we initially hand-labeled 200 examples under a generic, domain-agnostic e-commerce taxonomy. A trivial majority-class baseline scored 95% accuracy under that taxonomy — a strong signal one class dominated and the taxonomy did not differentiate AppleSupport’s actual traffic. We discarded it and relabeled the full golden set under the Apple-specific taxonomy above (Decision Log, item 8).

2.2 Golden Evaluation Set

We hand-labeled 200 examples. Customer messages were deduplicated (removing near-identical boilerplate), TF-IDF + KMeans ($k = 8$) was applied to the deduplicated pool, and examples were sampled roughly evenly across clusters (with a floor per cluster) so rare intents were not excluded by chance. Each example was hand-labeled for `gold_intent`, `gold_escalate` (yes/no), and a short `gold_escalate_reason`, via a resumable CLI labeling tool built for this project.

Intent	Count	% of set
software_bug_glitch	77	38.5%
system_update_issue	27	13.5%
other	23	11.5%
battery_performance	22	11.0%
device_hardware_connectivity	14	7.0%
app_store_media	14	7.0%
complaint_no_specific_ask	12	6.0%
apple_id_account	11	5.5%

Table 1: Golden set intent distribution (n=200)

`gold_escalate` distribution: 196 “no” / 4 “yes” (98%/2%). This severe imbalance is discussed directly in Section 5.

2.3 System Architecture

- **Retrieval:** TF-IDF vectorization + cosine similarity over $\sim 3,000$ historical pairs. Excludes a golden-set example’s own historical pair during evaluation to avoid trivial leakage.

- **Classification:** prompted LLM call (Groq, Llama 3.1 8B Instant) constrained to the eight taxonomy keys, temperature 0.
- **Drafting:** prompted LLM call grounded in the top-3 retrieved historical pairs, instructed not to invent policy details absent from those examples.
- **Routing:** deterministic, auditable rule-based logic (not LLM judgment) — escalates on (a) whole-word risk-keyword match, (b) policy-escalated intent (`apple_id.account`), or (c) low retrieval similarity (similarity < 0.12). Every decision returns a stated reason.

3 Results vs. Baselines

We compare the agent against two baselines on the same held-out test split (stratified 70/30 split of the golden set, $n = 140$ train / $n = 60$ test): a **trivial** baseline (always predicts the majority class) and a **simple** baseline (TF-IDF + Logistic Regression for intent; keyword-match rule for escalation).

Metric	Trivial	Simple	Agent
Intent accuracy	0.38	0.45	0.67
Intent F1 (macro)	0.07	0.34	0.69
Escalation accuracy	0.98	0.95	0.93

Table 2: Agent vs. baselines on held-out test split

The agent clearly and meaningfully outperforms both baselines on intent classification. Escalation accuracy is discussed in Section 5, since raw accuracy is a misleading metric on this severely imbalanced label.

3.1 Reply Quality (LLM-as-Judge)

Each drafted reply was scored 0–5 on three dimensions by an LLM judge (Groq gpt-oss-120b, temperature 0, strict rubric):

Dimension	Mean score (0–5)
Groundedness	4.87
Tone fit	4.70
Helpfulness	4.72

3.2 Judge ↔ Human Agreement

The author blind-scored 20 randomly sampled draft replies on the same three dimensions, without seeing the judge’s scores. Because both judge and human scores cluster near the top of the 0–5 scale, Pearson correlation is numerically unstable (undefined when a dimension has zero variance) — exact-match rate and mean absolute error (MAE) are the more reliable metrics here.

Groundedness and tone fit show strong agreement (MAE 0.20–0.25, within-1-point rates of 95–100%). Helpfulness shows materially weaker agreement (MAE 0.90, exact match only 20%), suggesting groundedness and tone-fit scores can be trusted as reported, while helpfulness scores should be treated as a softer signal pending rubric refinement.

Dimension	Exact match	Within 1 pt	Pearson corr.	MAE
Groundedness	0.90	0.95	−0.05*	0.20
Tone fit	0.75	1.00	undefined*	0.25
Helpfulness	0.20	0.95	−0.12*	0.90

Table 3: Judge $\leftrightarrow$ human agreement (n=20). *Correlation not treated as reliable evidence — see Section 5.

4 Failure Analysis — Top 5 Failure Modes

4.1 Keyword substring false positives (found and fixed)

Early escalation logic matched risk keywords via plain substring search. The keyword “sue” matched inside the word “issues,” causing benign technical complaints to escalate incorrectly. *Hypothesis*: naive substring matching is unsafe for short risk-keyword lists. *Fix*: switched to regex word-boundary matching (`\b` anchors). Escalation accuracy on the same test split improved from 0.90 to 0.93 immediately after the fix.

4.2 Blanket intent-level escalation over-triggers on benign account questions

The policy “always escalate `apple_id_account`” treats routine account-management questions the same as genuine security incidents. *Hypothesis*: this is a deliberate, stated safety-first tradeoff (Decision Log item 7), not a bug — the cost of missing a real account-hijack case is judged higher than the cost of unnecessary human review of routine questions.

4.3 Missed genuine escalation: no signal for repeated-failure / escalation fatigue

A customer reporting “4 senior techs couldn’t help fix my problem... want a refund” was not escalated. *Hypothesis*: the router has no mechanism to detect repeated-failure signals, only keyword and intent triggers — a real gap, since this is exactly the case that most needs a human.

4.4 Class imbalance limits reliability on rare intents

`software_bug_glitch` makes up 38.5% of the golden set; `apple_id_account` and `complaint_no_specific_ask` each have only 11–12 examples. *Hypothesis*: our sample size at the rare-class tail is too small to confirm the agent generalizes well there specifically, even though macro-F1 (0.69) is a fairer summary than accuracy given the skew.

4.5 LLM judge and human diverge specifically on “helpfulness”

MAE of 0.90 and only 20% exact match on helpfulness, versus much stronger agreement on groundedness and tone fit. *Hypothesis*: helpfulness is the most subjective of the three rubric dimensions — a holistic 0–5 judgment without an anchoring reference, unlike the more mechanically checkable groundedness and tone-fit dimensions.

5 What Is Misleading About My Headline Number?

- Escalation accuracy (93%) looks close to or worse than the trivial baseline (98%), but the gold label distribution is 196 “no” / 4 “yes.” A baseline that always predicts “no” scores 98% almost by definition, and our test split contains only 1 true positive — not enough to estimate recall reliably.

- The judge’s mean scores (4.7–4.87/5) look uniformly strong, but the human-agreement check shows this confidence is unevenly earned: groundedness and tone-fit agreement is strong, helpfulness agreement is weak (MAE 0.90). Averaging across dimensions hides this split.
- Pearson correlation values for judge/human agreement are unstable or undefined ($n=20$, scores clustered near ceiling) — a naive read would wrongly suggest stronger disagreement than the MAE/exact-match numbers actually show.
- Intent accuracy (67%) beats both baselines convincingly, but is computed on a set where one intent makes up 38.5% of examples — a model disproportionately good at that one class can inflate this number without being reliably good at rarer, possibly more operationally important classes (e.g. `apple_id.account`, $n = 11$).
- *Grounded is not the same as correct.* Retrieval-and-draft guarantees textual similarity to a real historical exchange, not that the advice is actually the right fix for this specific customer’s device/iOS version.
- The golden set ($n = 200$) and test split ($n = 60$) are small enough that single-example flips can swing percentage-point results — several headline numbers should be read as directional signals from a promising early prototype, not statistically tight estimates.

6 What We’d Do With One More Week

- Expand the golden set specifically at the `escalate=yes` tail (currently 4/200) so escalation recall can be measured reliably.
- Replace the blanket `apple_id.account` escalation rule with a hybrid policy: escalate only when the intent also co-occurs with a risk keyword or low retrieval confidence.
- Add an explicit “repeated failure / escalation fatigue” signal to the router (e.g. phrases like “already tried,” “still not fixed,” or a lightweight frustration score).
- Tighten the LLM judge’s helpfulness rubric — e.g. require it to quote the specific sentence it considers the “next step” — and re-run the human-agreement check.
- Run k -fold cross-validation (not a single 70/30 split) for baseline comparisons, since a single split on a moderately small golden set is sensitive to which rows land in test.
- Add a per-intent precision/recall breakdown table, not just macro-F1, so rare-class reliability is visible rather than averaged away.
- Run an ablation comparing TF-IDF retrieval against sentence-embedding retrieval to quantify what was given up by the dependency-lightweight TF-IDF choice.

Appendix: Reproducibility

Full setup and run instructions are provided in `README.md` in the project repository. The pipeline reproduces all headline results above in under 15 minutes on a subsample, per the assignment’s guidance that a subsample is expected and encouraged.